

CAP5510 – Bioinformatics

Fall 2025

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University of Florida

Vital Information

- Instructor: Tamer Kahveci
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- Office: MH 5400C
- Time: M/W/F (10:40-11:30AM)
- Office hours: M/F 11:50-12:40 AM
- Course page:
 - <http://www.cise.ufl.edu/~tamer/teaching/fall2025>

Goals

- Understand the major components of bioinformatics data and how computer technology is used to understand this data better.
- Learn main potential research problems in bioinformatics and gain background information.

This Course will not

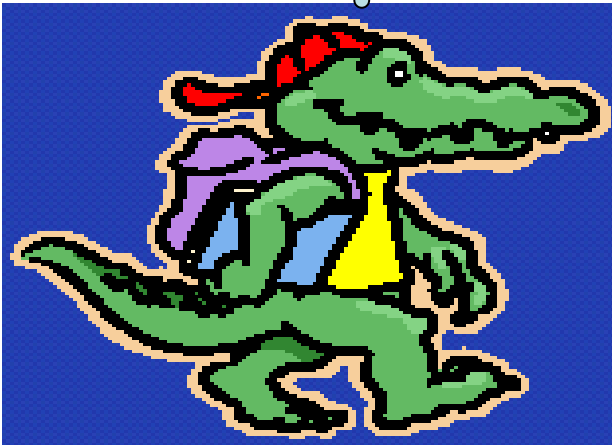
- Teach you biology.
- Teach you programming
- Teach you how to be an expert user of off-the-shelf molecular biology computer packages.

Course Outline

- Introduction to terminology
- Biological sequences
- Sequence comparison
 - Lossless alignment (DP)
 - Lossy alignments (BLAST, etc)
- Sequence assembly
- Substitution matrices, statistics
- Multiple sequence alignment
- Phylogeny
- Protein structures and their prediction
- Biological networks

Grading

How can I get an A?



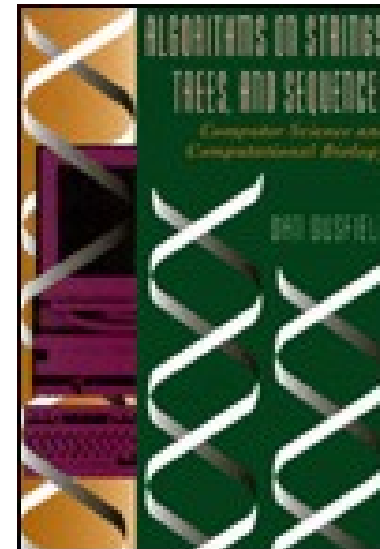
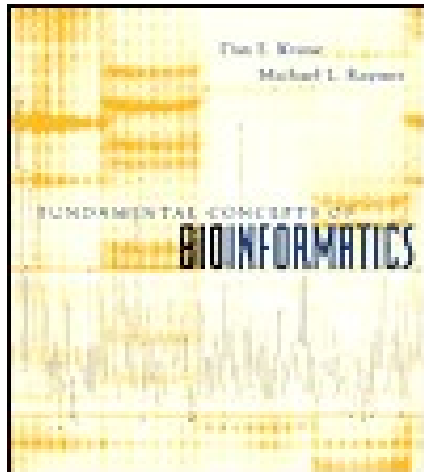
1. Project (50 %)
 - Contribution (2.5 % bonus)
2. Other (50 %)
 - Homeworks + quizzes

Expectations

- Background needed
 - Data structures and algorithms.
 - Coding (C, Java, etc)
- Encourage
 - read bioinformatics literature in general

Text Book

- No text book.
- Class notes + papers.



Where to Look ?

- Journals
 - Bioinformatics
 - Genome Research
 - PLOS Computational Biology
 - Journal of Computational Biology
 - IEEE Transaction on Computational Biology and Bioinformatics
- Conferences
 - RECOMB
 - ISMB
 - ECCB
 - PSB
 - BCB

What is Bioinformatics?

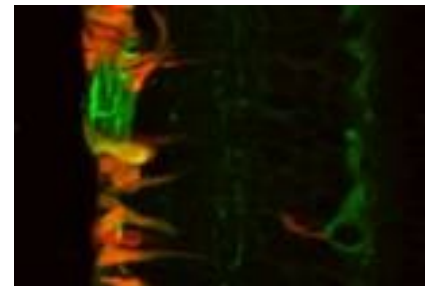
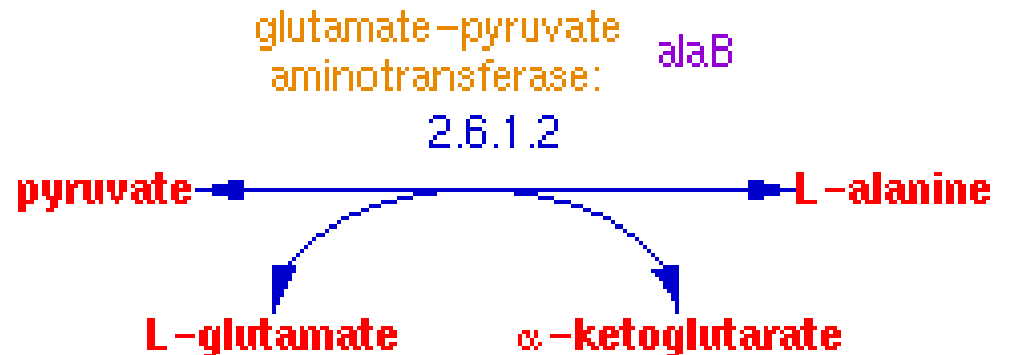
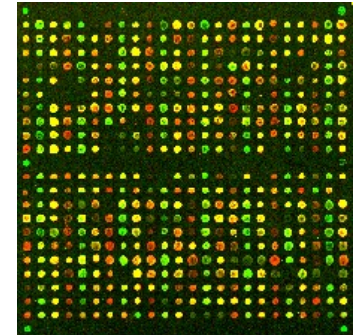
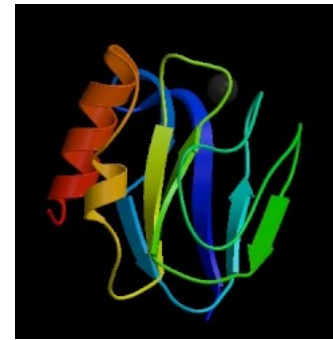
- Bioinformatics is the field of science in which biology, computer science, and information technology merge into a single discipline. The ultimate goal of the field is to enable the discovery of new biological insights as well as to create a global perspective from which unifying principles in biology can be discerned. There are three important sub-disciplines within bioinformatics:
 - the development and implementation of tools that enable efficient access and management of different types of information.
 - the analysis and interpretation of various types of data including nucleotide and amino acid sequences, protein domains, and protein structures
 - the development of new algorithms and statistics with which to assess relationships among members of large data sets

From NCBI (National Center for Biotechnology Information)
<http://www.ncbi.nlm.nih.gov/Education/BLASTinfo/milestones.html>

Does biology have anything to do with computer science?

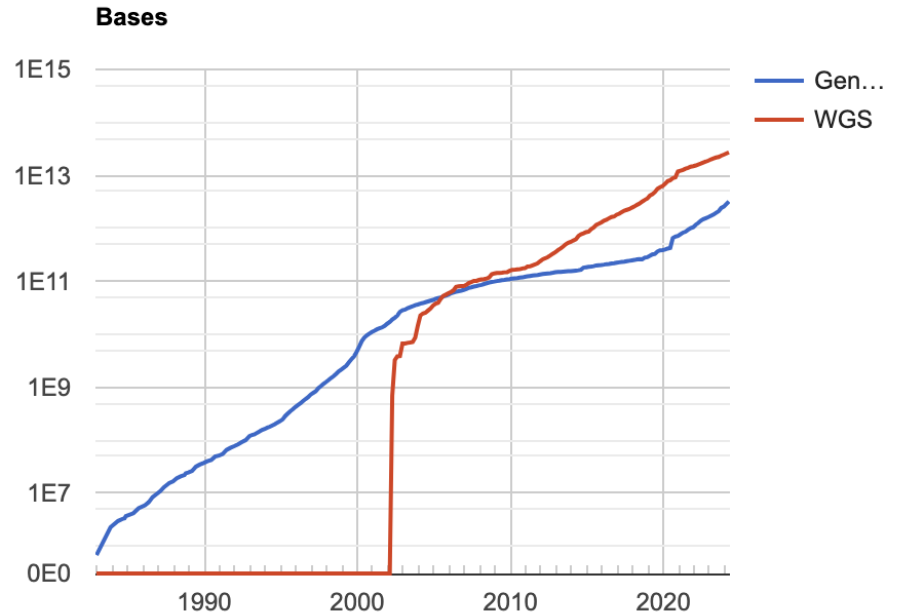
Challenges 1/4

- Data diversity
 - DNA
(ATCCAGAGCAG)
 - Protein sequences
(MHPKVDALLSR)
 - Protein structures
 - Microarrays
 - Biological networks
 - Bio-images
 - Time series



Challenges 2/4

- Database size as of April 2024,
 - GeneBank : More than 30×10^{13} bases.
 - More than 500K protein sequences,
 - More than 214K protein structures in PDB



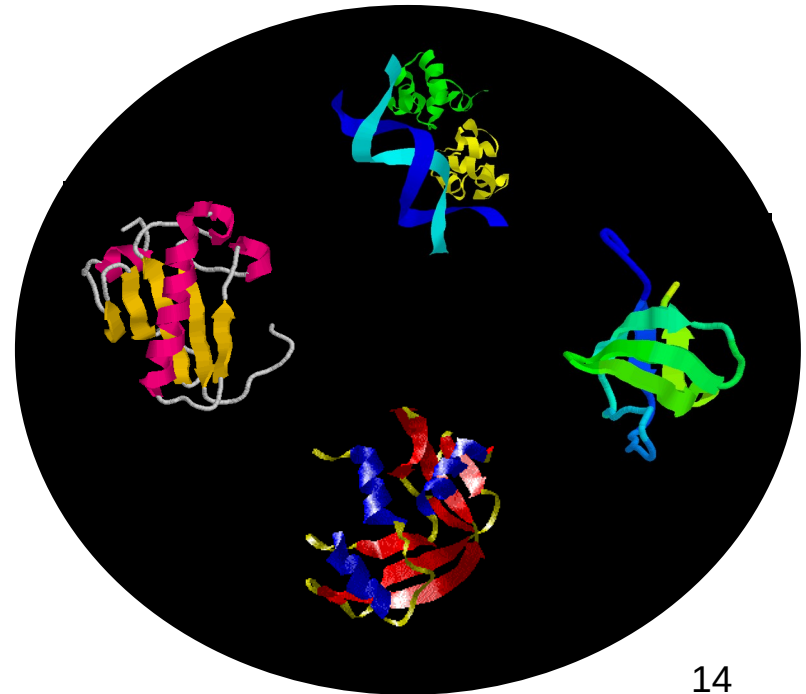
Genome sequence now accumulate so quickly that, in less than a week, a single laboratory can produce more bits of data than Shakespeare managed in a lifetime, although the latter make better reading.

Challenges 3/4

- Deciphering the code
 - Within same data type: hard
 - Across data types: harder

```
caacaagccaaaactcgtacaaatatgaccgcacttcgctataaagaacacggcttgtgg  
cgagatatctcttggaaaaactttcaagagcaactcaatcaactttctcgagcattgctt  
gctcacaatatattgacgtacaagataaaatcgccatttttgcccataatatggaacgttg  
gttggtcatgaaactttcgggtatcaaagatgggttaatgaccactgttcacgcaacgact  
acaatcgttgacattgcgaccttacaaattcgagcaatcacagtgccattttacgcaacc  
aatacagcccagcaagcagaatttatcctaaatcacgccgatgtaaaaattctcttcgctc  
ggcgatcaagagcaatacgaatcaaacattggaaattgctcatcattgtccaaaattacaa  
aaaattgtagcaatgaaatccaccattcaattacaacaagatcctcttcttgcaacttg
```

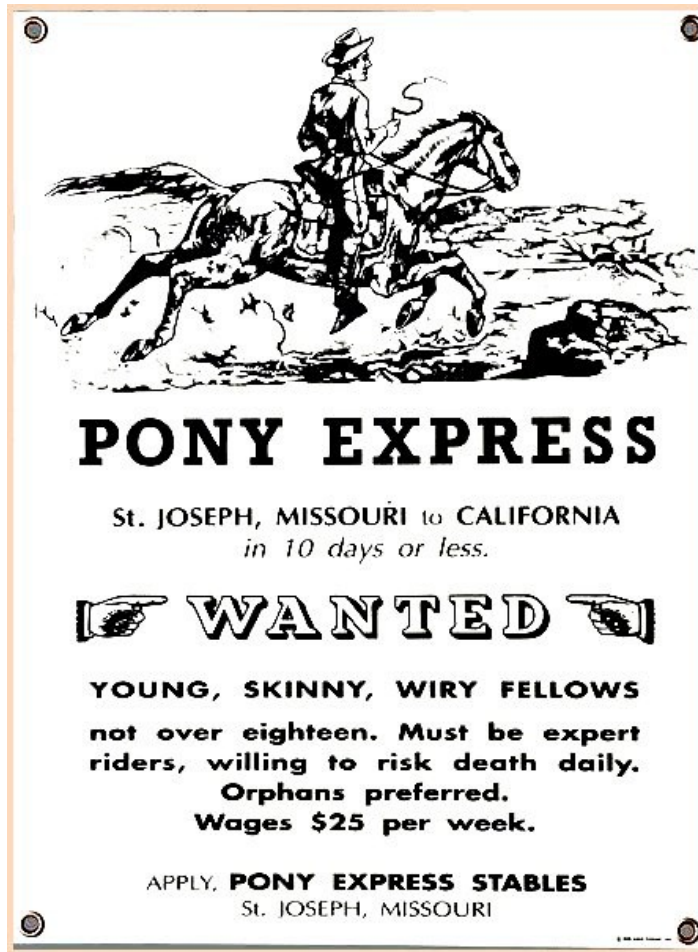
```
atggcaattaaaattggtatcaatgggttttggtcgtatcgggccgtatcgtattccgtgca  
gcacaacaccgtgatgacattgaagtgttaggtattaacgacttaatcgacgttgaatac  
atggcttatatgttgaaatatgattcaactcacggtcgtttcgacggcactgttgaagtg  
aaagatggtaacttagtggttaatggtaaaactatccgtgtaactgcagaacgtgatcca  
gcaaaactaaactggggtgcaatcggtgttgatatacgtgttggaagcgactgggttatctc  
ttaactgatgaaactgctcgtaaacatatcactgcaggcgcaaaaaaagttgtattaact  
ggccccatcaaagatgcaaccctatgttcgttcgtggtgtaaaacttcaacgcatacga  
ggtcaagatatcgtttctaacgcattctgtacaacaaactgtttagctccttttagcacgt  
gttggtcatgaaactttcgggtatcaaagatgggttaatgaccactgttcacgcaacgact
```



Challenges 4/4

- Inaccuracy
- Redundancy

What is the Real Solution?



We need better computational methods

- Compact summarization
- Fast and accurate analysis of data
- Efficient indexing

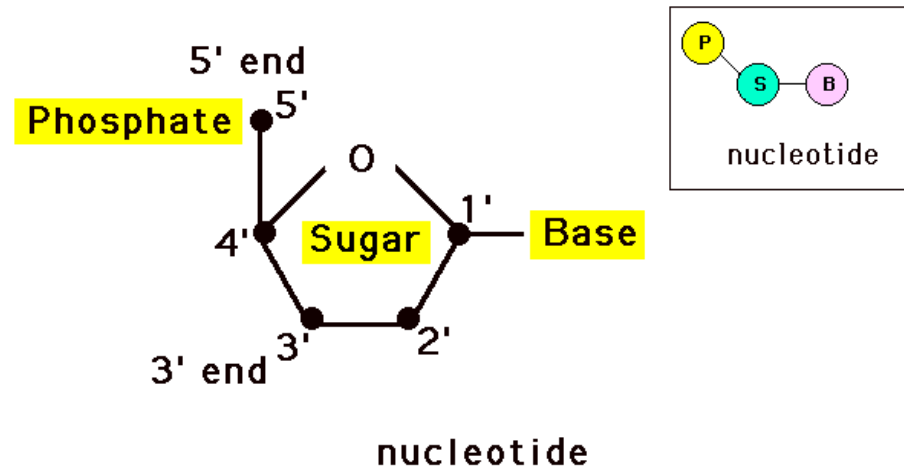
A Gentle Introduction to Molecular Biology

Goals

- Understand major components of biological data
 - DNA, protein sequences, expression arrays, protein structures
- Get familiar with basic terminology
- Learn commonly used data formats

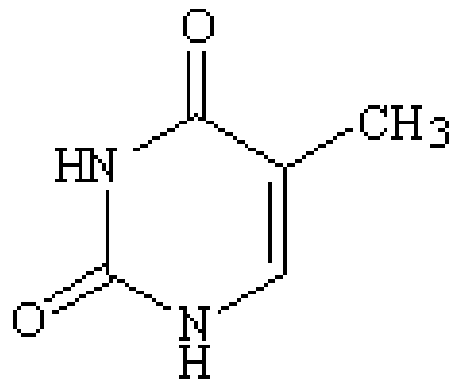
Genetic Material: DNA

- Deoxyribonucleic Acid, 1950s
 - Basis of inheritance
 - Eye color, hair color, ...
- 4 nucleotides
 - A, C, G, T

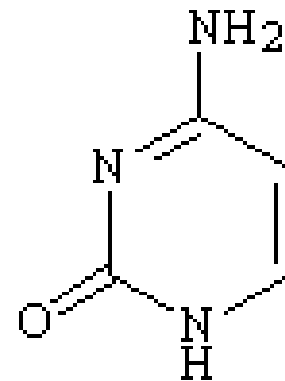


Chemical Structure of Nucleotides

Pyrimidines

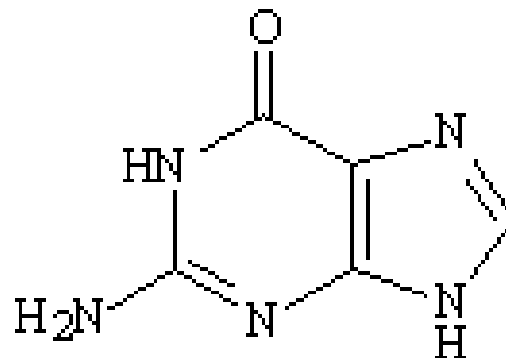


Thymine

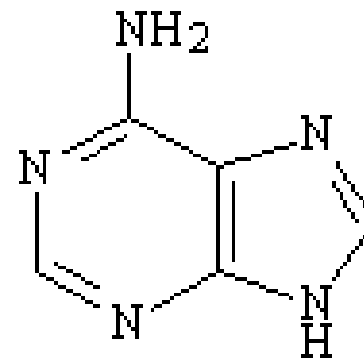


Cytosine

Purines

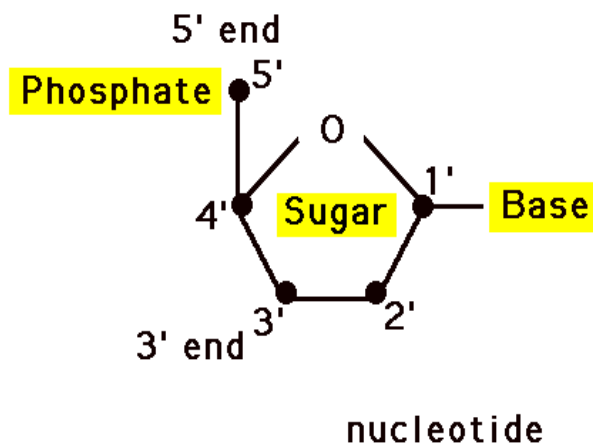


Guanine

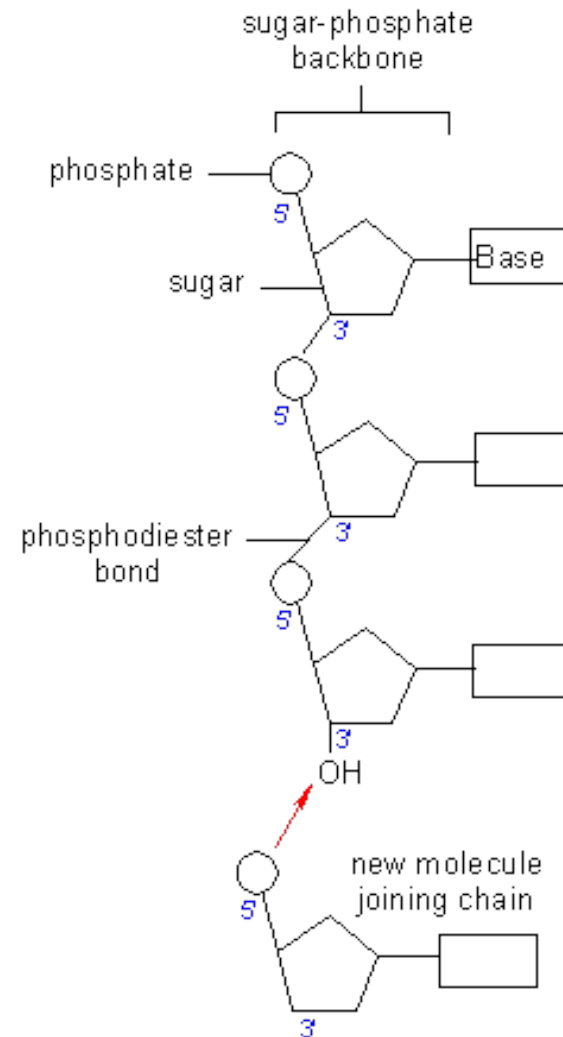


Adenine

Making of Long Chains

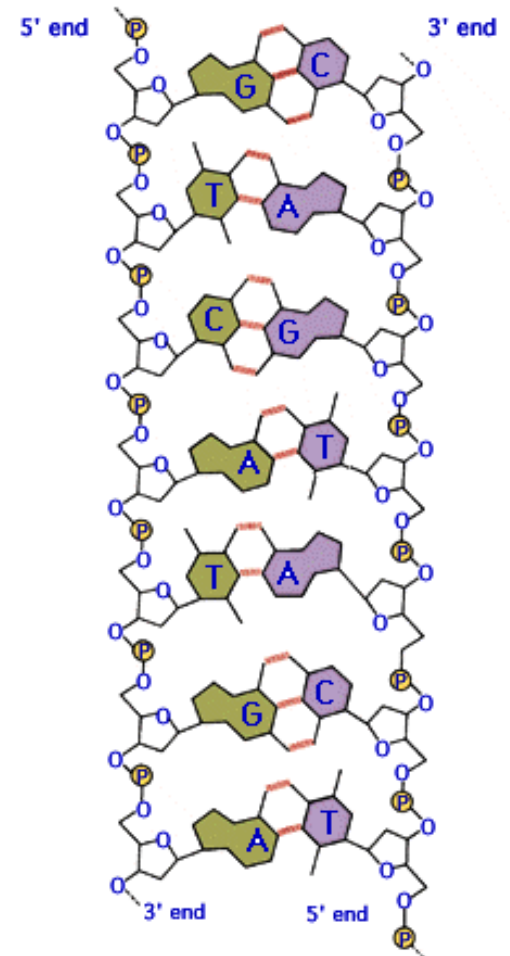
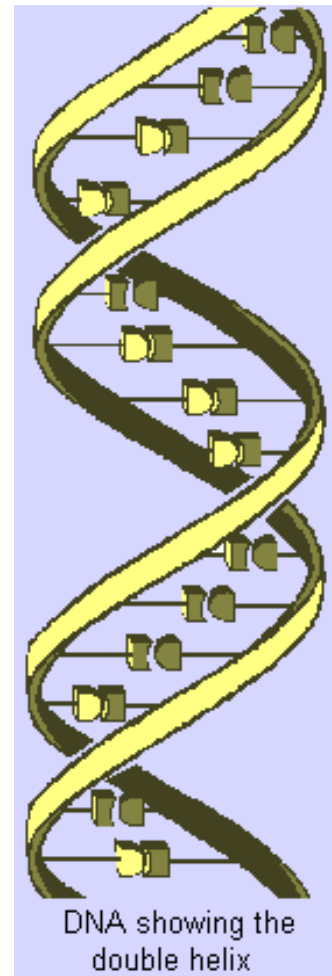


5' → 3'



DNA structure

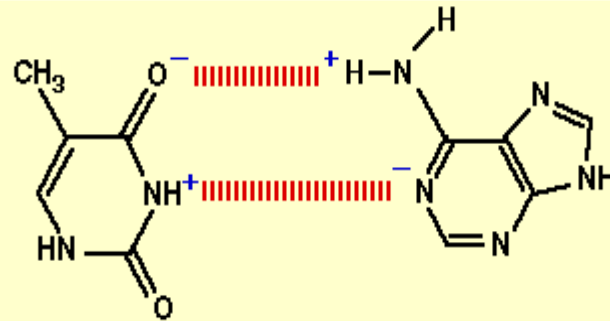
- Double stranded, helix (Watson & Crick)
- Complementary
 - A-T
 - G-C
- Antiparallel
 - 3' -> 5' (downstream)
 - 5' -> 3' (upstream)



Base Pairs

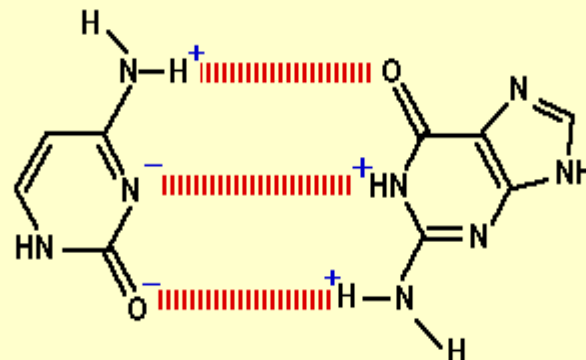
PYRIMIDINES

PURINES



Thymine

Adenine



Cytosine

Guanine

Question

- 5' - GTTACA – 3'
- 5' – XXXXXX – 3' ?
- 5' – TGTAAC – 3'
- Reverse complements.

Repetitive DNA

- Tandem repeats: highly repetitive
 - Satellites (100 k – 1 Gbp) / (a few hundred bp)
 - Mini satellites (1 k – 20 kbp) / (9 – 80 bp)
 - Micro satellites (< 150 bp) / (1 – 6 bp)
 - DNA fingerprinting
- Interspersed repeats: moderately repetitive
 - LINE
 - SINE
- Proteins contain repetitive patterns too

Genetic Material: an Analogy

- Nucleotide => letter
- Gene => sentence
- Contig => chapter
- Chromosome => book
 - Traits: Gender, hair/eye color, ...
 - Disorders: down syndrome, turner syndrome, ...
 - Chromosome number varies for species
 - We have 46 (23 + 23) chromosomes
- Complete genome => volumes of encyclopedia
- Hershey & Chase experiment show that DNA is the genetic material. ([ch14](#))

Functions of Genes 1/2

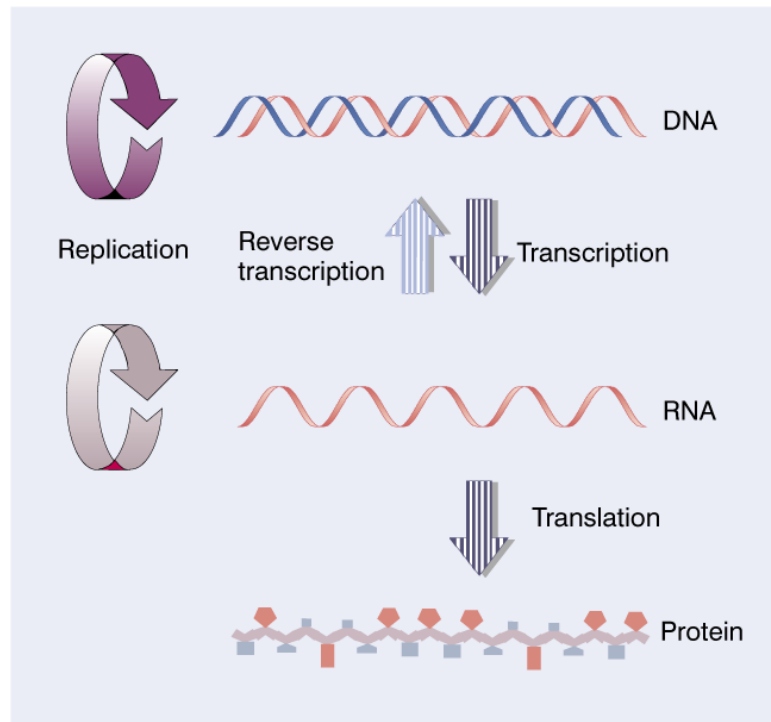
- **Signal transduction:** sensing a physical signal and turning into a chemical signal
- **Enzymatic catalysis:** accelerating chemical transformations otherwise too slow.
- **Transport:** getting things into and out of separated compartments

Functions of Genes 2/2

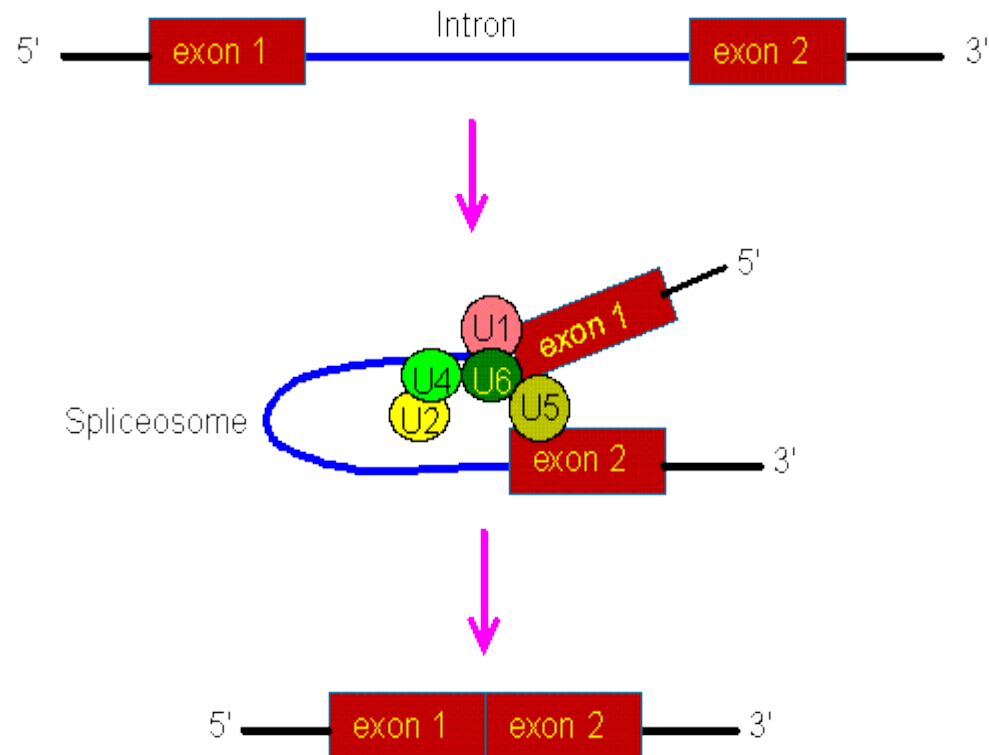
- **Movement:** contracting in order to pull things together or push things apart.
- **Transcription control:** deciding when other genes should be turned ON/OFF
 - Animation ([ch7](#))
- **Structural support:** creating the shape and pliability of a cell or set of cells

Central Dogma

Figure 1.34 The central dogma states that information in nucleic acid can be perpetuated or transferred, but the transfer of information into protein is irreversible.



Introns and Exons 1/2

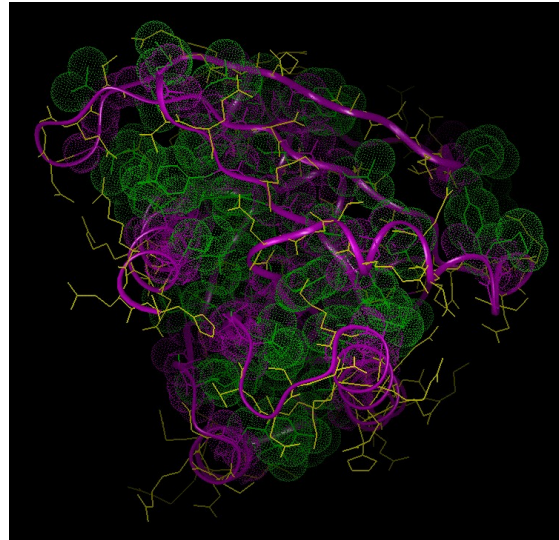
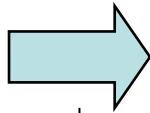


Introns and Exons 2/2

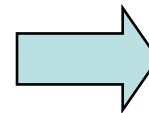
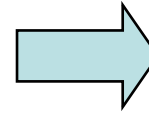
- Humans have about 20,000 genes < 3% of total DNA in genome.
- Remaining 97% for control information. (e.g. when, where, how long, etc...)



DNA
(Genotype)



Protein



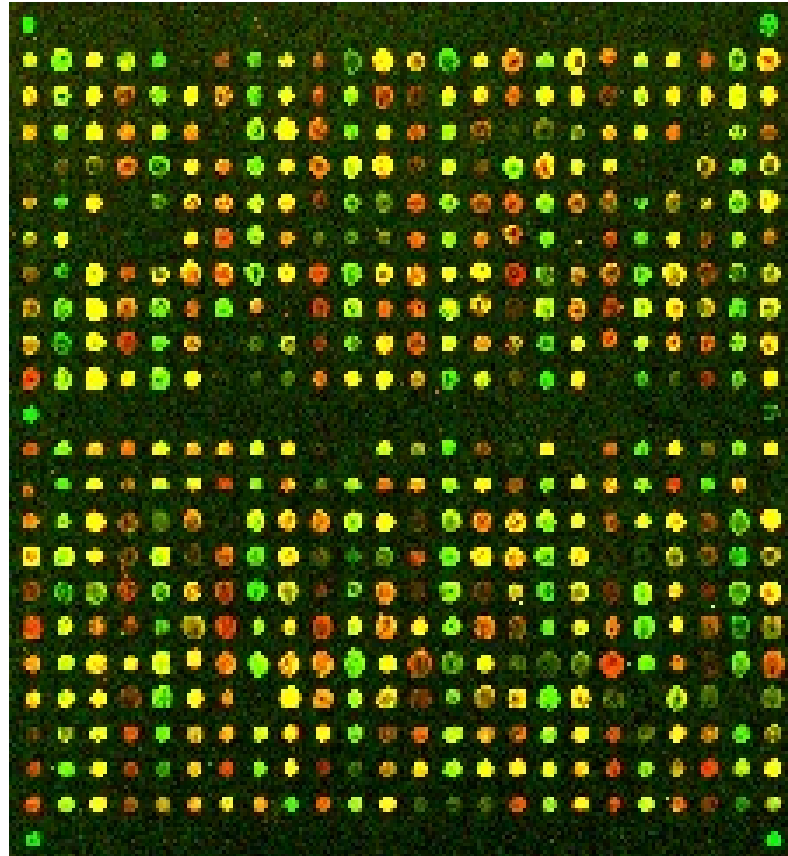
Phenotyp
e

Gene expression

Gene Expression

- Building proteins from DNA
 - Promoter sequence: start of a gene
 - ≥ 13 nucleotides.
- Positive regulation: proteins that bind to DNA near promoter sequences increases transcription.
- Negative regulation

Microarray



[Animation on creating microarrays](#)

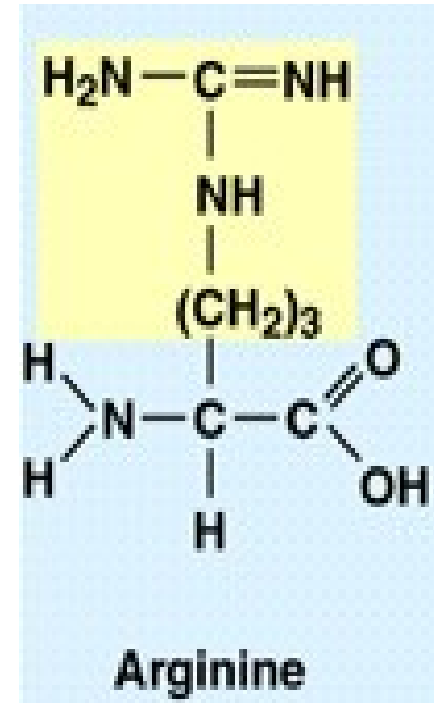
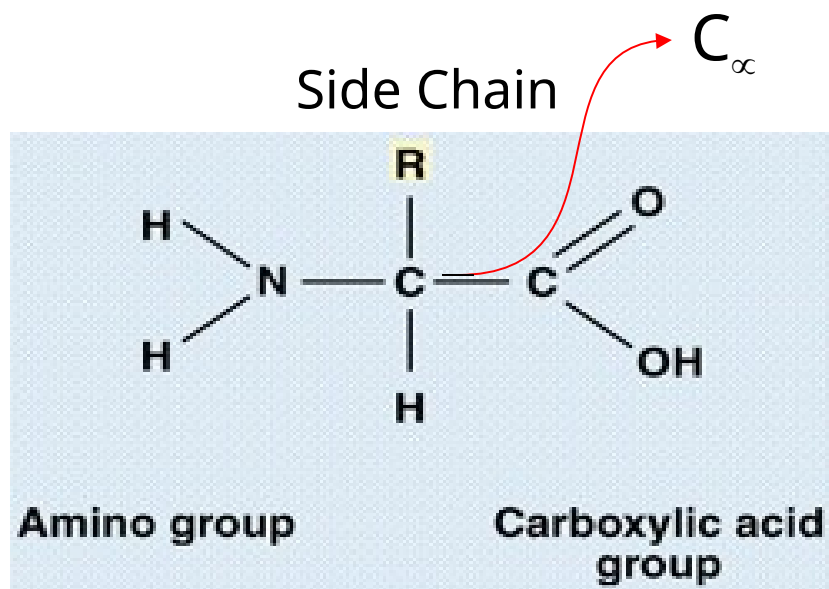
Amino Acids

- 20 different amino acids
 - ACDEFGHIKLMNPQRSTVWY but not BJOUXZ
- ~300 amino acids in an average protein, hundreds of thousands known protein sequences
- How many nucleotides can encode one amino acid ?
 - $4^2 < 20 < 4^3$
 - E.g., Q (glutamine) = CAG
 - degeneracy
 - Triplet code (codon)

Triplet Code

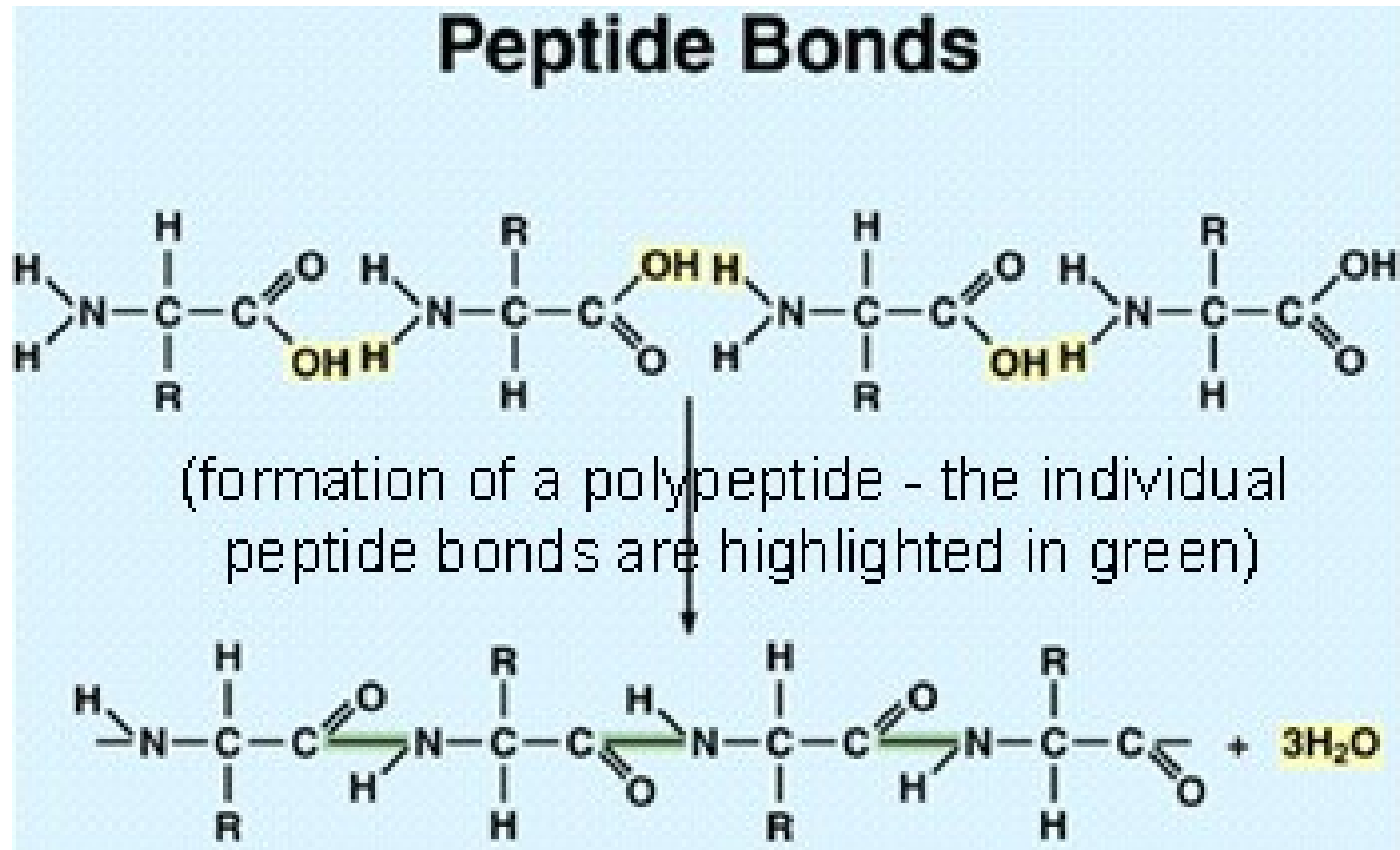
One-letter code	Amino acid	Three-letter code	Genetic code
A	Alanine	Ala	GC*
C	Cysteine	Cys	UGU, UGC
D	Aspartic Acid	Asp	GAU, GAC
E	Glutamic Acid	Glu	GAA, GAG
F	Phenylalanine	Phe	UUU, UUC
G	Glycine	Gly	GG*
H	Histidine	His	CAU, CAC
I	Isoleucine	Ile	AUU, AUC, AUA
K	Lysine	Lys	AAA, AAG
L	Leucine	Leu	UUA, UUG, CU*
M	Methionine	Met	AUG
N	Asparagine	Asn	AAU, AAC
P	Proline	Pro	CC*
Q	Glutamine	Gln	CAA, CAG
R	Arginine	Arg	CG*, AGA, AGG
S	Serine	Ser	UC*, AGU, AGC
T	Threonine	Thr	AC*
V	Valine	Val	GU*
W	Tryptophan	Trp	UGG
Y	Tyrosine	Tyr	UAU, UAC

Molecular Structure of Amino Acid

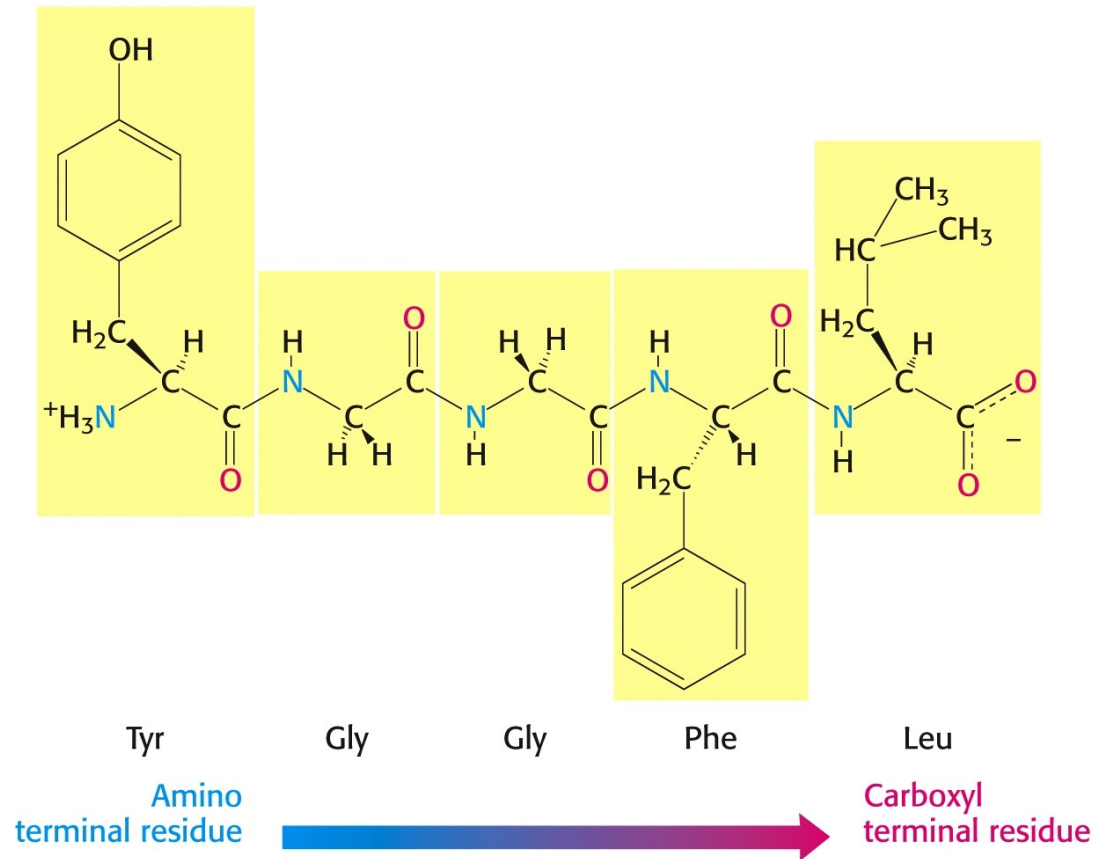


- Non-polar, Hydrophobic (G, A, V, L, I, M, F, W, P)
- Polar, Hydrophilic (S, T, C, Y, N, Q)
- Electrically charged (D, E, K, R, H)

Peptide Bonds



Direction of Protein Sequence



Animation on protein synthesis ([ch15](#))

Data Format

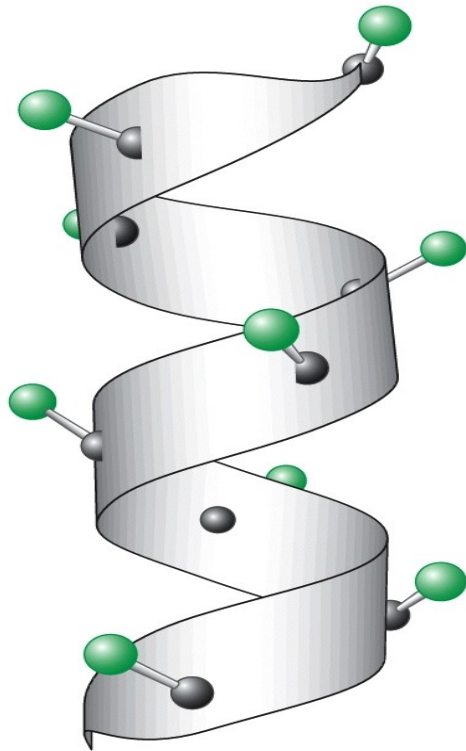
- [GenBank](#)
- [EMBL](#) (European Mol. Biol. Lab.)
- [SwissProt](#)
- FASTA
- [NBRF](#) (Nat. Biomedical Res. Foundation)
- Others
 - IG, GCG, Codata, ASN, GDE, Plain ASCII

Primary Structure of Proteins

>2IC8:A|PDBID|CHAIN|SEQUENCE

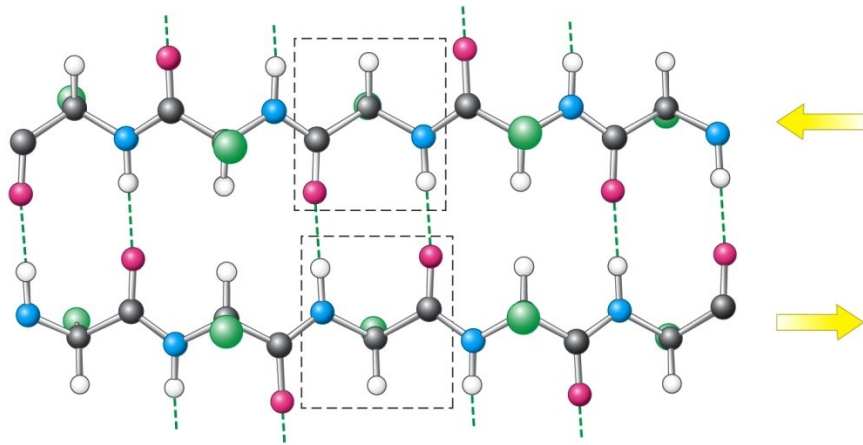
ERAGPVTWVMMIACVVVFIAMQILGDQ
EVMLWLAWPFDPTLKFEFWRYFTHALM
HFSLMHILFNLLWWWYLGGAWEKRLGS
GKLIVITLISALLSGYVQQKFSGPWFGGL
SGVVYALMGYVWLRGERDPQSGIYLQR
GLIIFALIWIWVAGWFDLFGMSMANGAHI
AGLAVGLAMAFVDSLNA

Secondary Structure: Alpha Helix

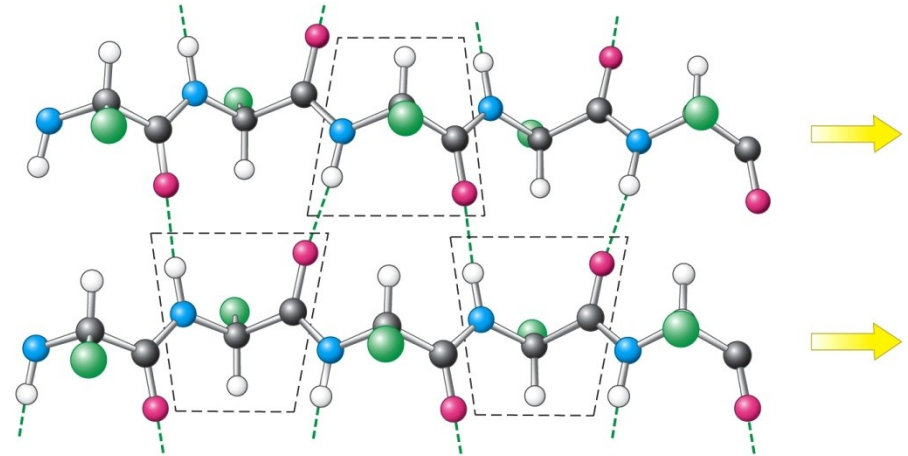


- 1.5 Å translation
- 100 degree rotation
- $\Phi = -60^\circ$
- $\Psi = -60^\circ$

Secondary Structure: Beta sheet



anti-parallel



parallel

$\Phi = -135$

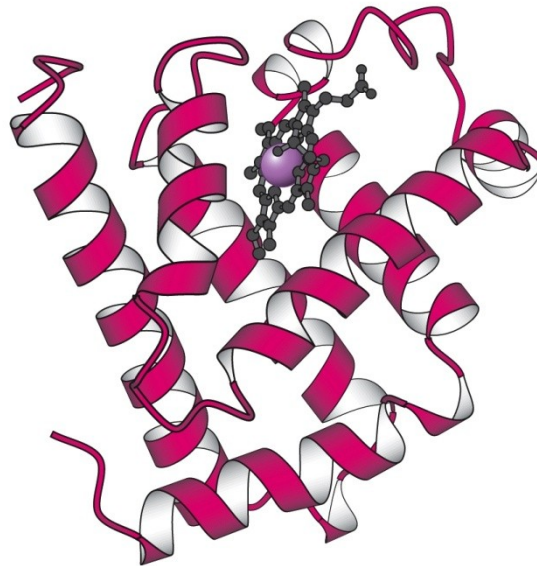
$\Psi = 135$

Tertiary Structure



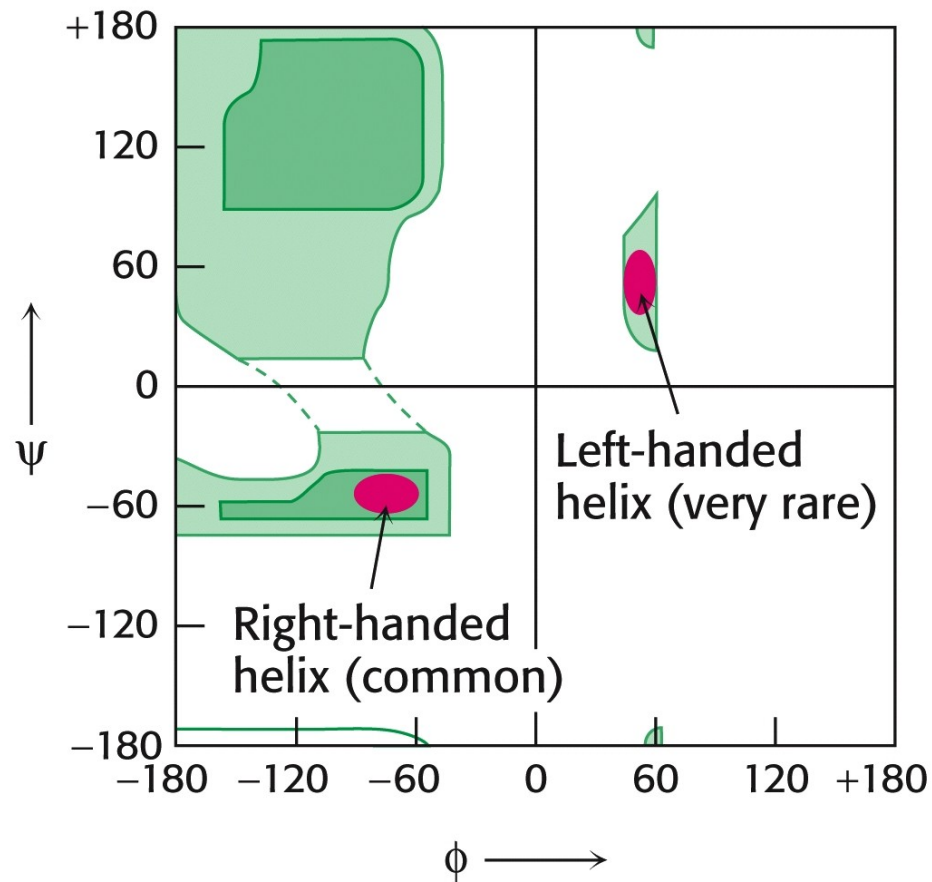
Tertiary Structure

- 3-d structure of a polypeptide sequence
 - interactions between non-local atoms



tertiary structure of
myoglobin

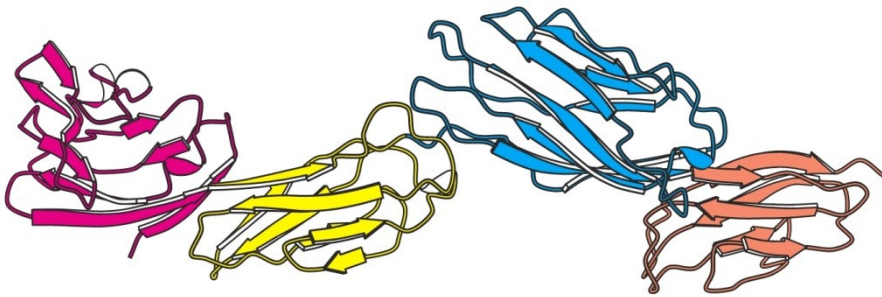
Ramachandran Plot



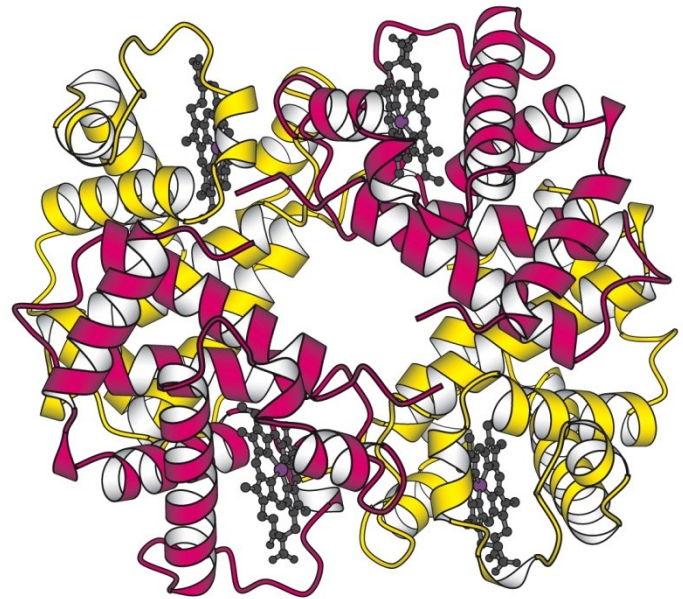
Sample pdb entry (<http://www.rcsb.org/pdb/>)

Quaternary Structure

- Arrangement of protein subunits



quaternary structure
of Cro



human hemoglobin
tetramer

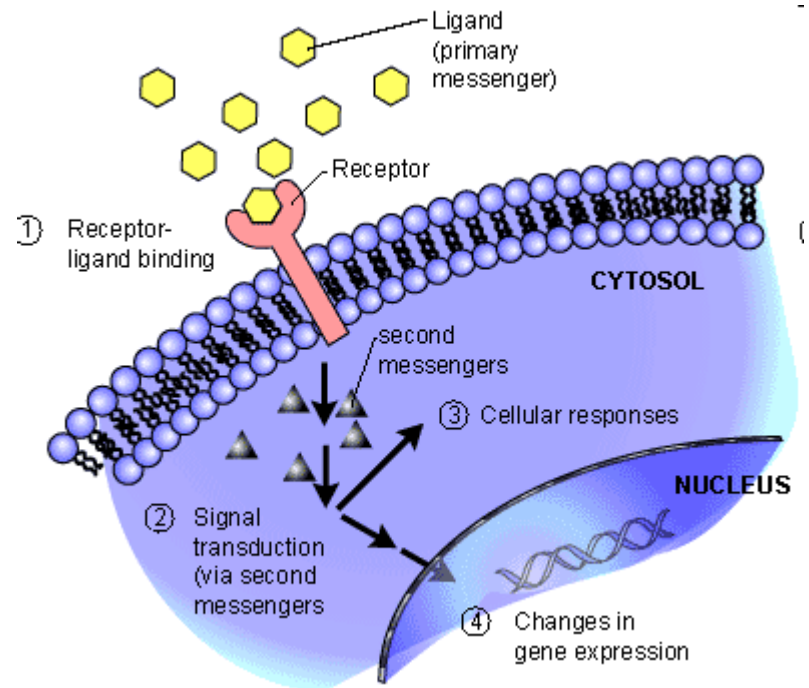
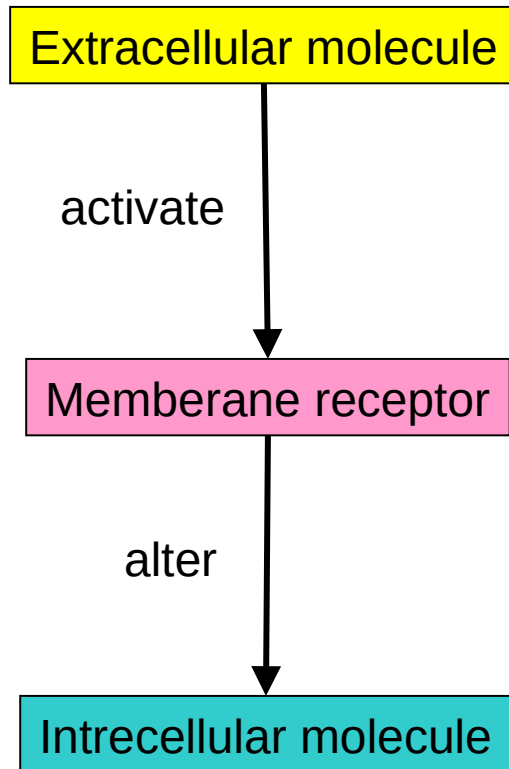
Structure Summary

- 3-d structure determined by protein sequence
- Prediction remains a challenge
- Diseases caused by misfolded proteins
 - Mad cow disease
- Classification of protein structure

Biological networks

- Signal transduction network
- Transcription control network
- Post-transcriptional regulation network
- PPI (protein-protein interaction) network
- Metabolic network

Signal transduction

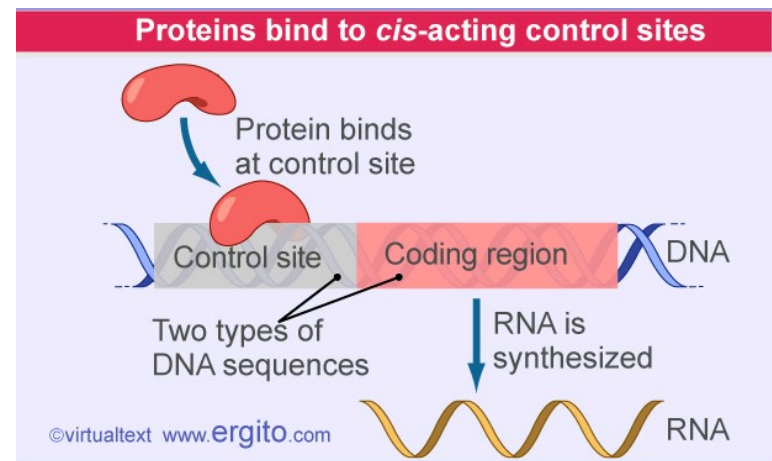


Transcription control network

Transcription Factor (TF) – some protein

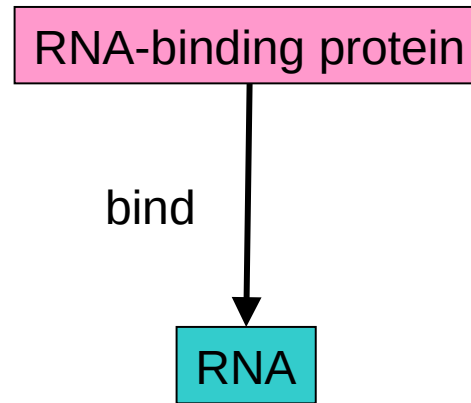
bind

Promoter region of a gene



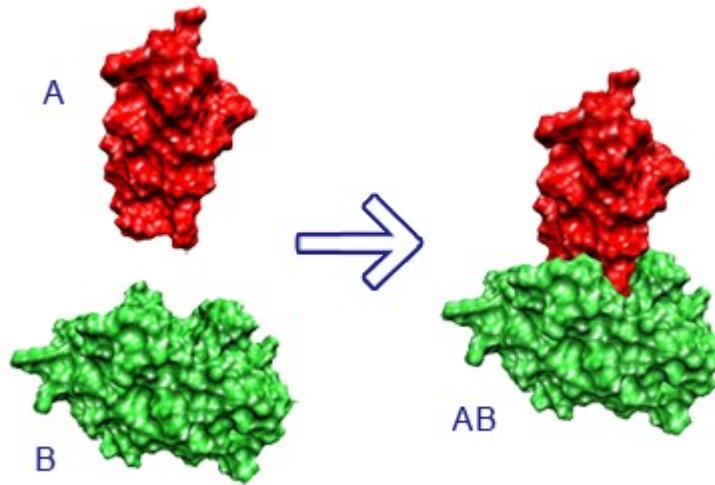
- Up/down regulates
- TFs are potential drug targets

Post transcriptional regulation



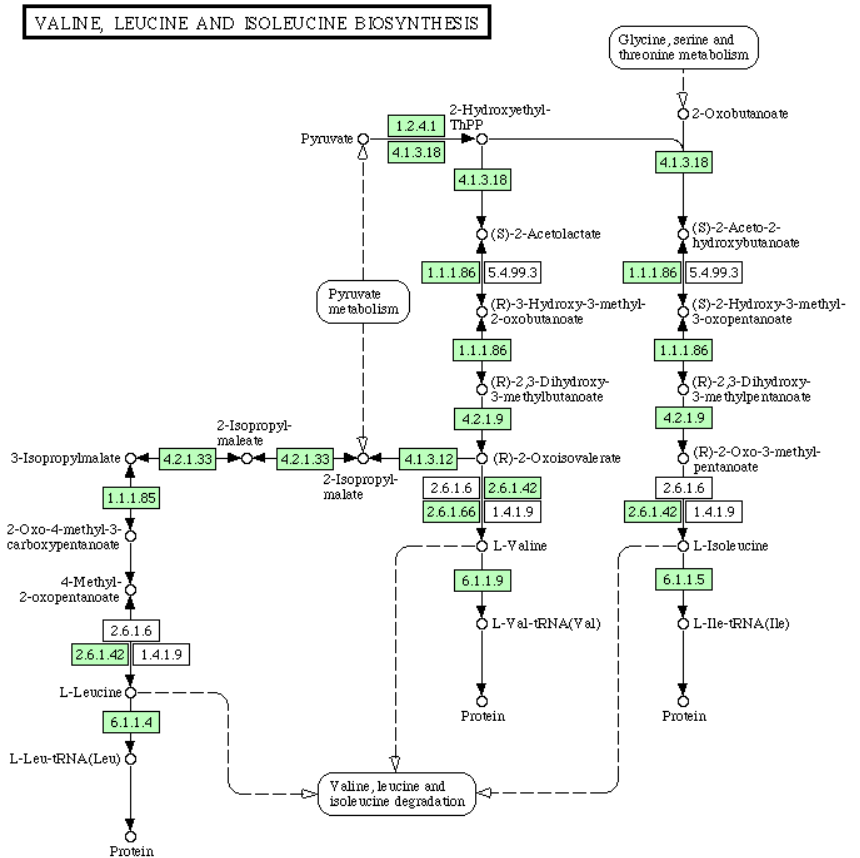
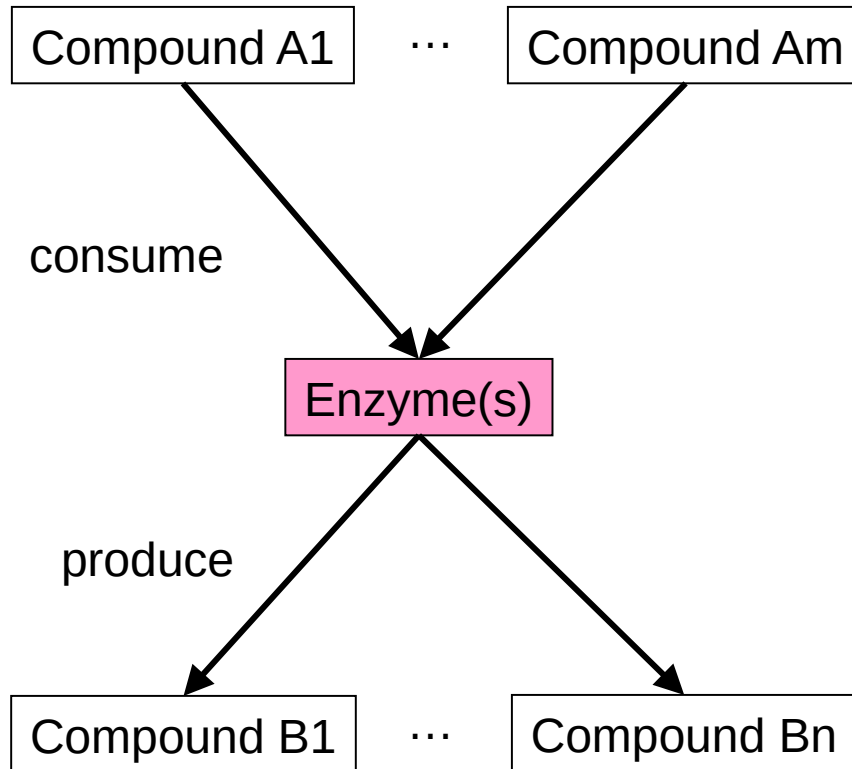
Slow down or accelerate protein translation from RNA

PPI (protein-protein interaction)



Creates a protein complex

Metabolic interactions





Quiz Next Lecture

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QUIZ

STOP

Next:

- Basic sequence comparison
- Dynamic programming methods
 - Global/local alignment
 - Gaps